

Tutorial Sheet 2

1. The surface source method is used to calculate the flow around a symmetric aerofoil section. Elements t and $t+1$ are located either side of the trailing edge, and the computer print-out gives the following results for the tangential velocities at their control points:

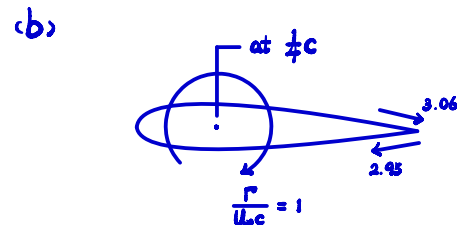
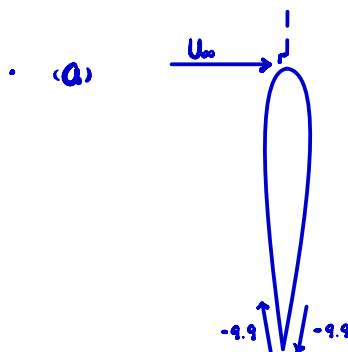
 V_θ

Onset flow

	t	$t+1$
(a) Unit free-stream velocity, aerofoil at 90° incidence	-9.9	-9.9
(b) <u>Unit circulation</u> $\frac{\Gamma}{U_\infty c} = 1$	3.06	2.95

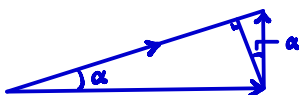
Calculate the lift coefficient at 10° incidence. In a real fluid, would you expect the C_L to be greater or less than this value?

ANS. 1.14



To enforce the Kutta-condition, $u_u - u_l = 0$ at T.E.

$$\Delta Q_t + k \Delta Q_{t,r} = 0$$



$$\text{where } \Delta Q_t = \Delta Q_t|_{\alpha=0} \cos \alpha + \Delta Q_t|_{\alpha=90} \sin \alpha$$

$$= 0 + (-9.9 - 9.9) \sin 10^\circ = -3.438$$

$$\text{and } \Delta Q_{t,r} = 3.06 + 2.95 = 6.01$$

$$\therefore -3.438k + 6.01 = 0 \quad \therefore k = 0.572$$

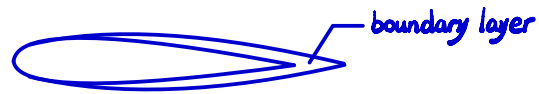
$$\therefore C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 c} = 2k = 1.144$$

- In real fluid, C_L going to be less than 1.144 due to loss of camber caused by growing boundary layer

• inviscid

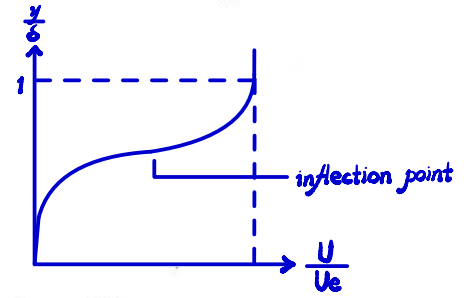


• real



2. The velocity profile in a laminar boundary layer over a smooth surface is given approximately by the expression:

$$U/U_e = 3(y/\delta)^2 - 2(y/\delta)^3$$



Show that the boundary layer is on the point of separating.

If the flow is on the verge of separation, will the pressure on the wall be increasing or decreasing with distance downstream? How would the separation point be affected if the boundary layer were turbulent?

$$\frac{\theta}{\delta} = \int_0^1 \frac{U}{U_e} \left(1 - \frac{U}{U_e}\right) d\left(\frac{y}{\delta}\right)$$

$$\text{Let } u = \frac{y}{\delta}$$

$$\begin{aligned} \therefore \frac{\theta}{\delta} &= \int_0^1 (3u^2 - 2u^3)(1 - 3u^2 + 2u^3) du \\ &= \int_0^1 (3u^2 - 9u^4 + 6u^5 - 2u^3 + 6u^5 - 4u^6) du \\ &= \int_0^1 (-4u^6 + 12u^5 - 9u^4 - 2u^3 + 3u^2) du \\ &= \left[-\frac{4}{7}u^7 + 2u^6 - \frac{9}{5}u^5 - \frac{1}{2}u^4 + u^3 \right]_0^1 \\ &= \frac{9}{70} \end{aligned}$$

$$\frac{\partial U}{\partial y^2} = U_e \frac{d^2}{dy^2} \left(\frac{3}{\delta^2} y^2 - \frac{2}{\delta^3} y^3 \right) = U_e \frac{d}{dy} \left(\frac{6}{\delta^2} y - \frac{6}{\delta^3} y^2 \right) = U_e \left(\frac{6}{\delta^2} - \frac{12}{\delta^3} y \right)$$

$$\therefore \frac{\partial U}{\partial y^2} \Big|_w = 6 \frac{U_e}{\delta^2} \text{ where } y = 0$$

$$\therefore m = \frac{\theta^2}{U_e} \left(\frac{\partial U}{\partial y^2} \right)_w = \frac{81}{4900} \delta^2 \cdot \frac{1}{U_e} \cdot 6 \frac{U_e}{\delta^2} = \frac{243}{2450} = 0.0992$$

- On the verge of separation, $\frac{dU}{dy} \Big|_w = 0$ and $\frac{d^2U}{dy^2} \Big|_{\text{inflection}} = 0$ while $0 < y_{\text{inflection}} < \delta$

$$\frac{dU}{dy} = U_e \left(\frac{6}{\delta^2} y - \frac{6}{\delta^3} y^2 \right)$$

$$\therefore \frac{dU}{dy} \Big|_w = \frac{dU}{dy} \Big|_{y=0} = 0$$

At inflection point,

$$\frac{d^2 U}{dy^2} \Big|_{y_{inf}} = U_e \left(\frac{6}{\delta^2} - \frac{12}{\delta^2} y \right) = 0 \quad \therefore \left(\frac{y}{\delta} \right)_{inf} = 0.5 \quad \therefore 0 < y_{inf} < \delta$$

- At wall.

$$\frac{dp}{dx} = \mu \frac{d^2 U}{dy^2} \Big|_w = 6 \frac{U_e}{\delta^2} > 0 \quad \therefore \text{adverse pressure gradient}$$

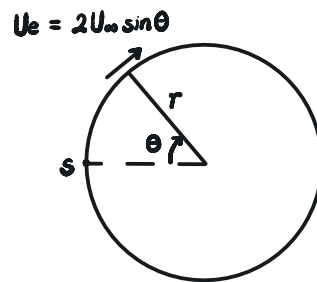
- Turbulence boundary layer always separate later than laminar one

3. The inviscid flow solution for the velocity around the surface of a circular cylinder in a cross flow with a velocity U_∞ is $U_e = 2U_\infty \sin \theta$, where θ is measured from the front stagnation point. Using Thwaites' method, find the separation angle for a laminar boundary layer growing on the cylinder.

At a sub-critical Reynolds number, experiments show that the laminar boundary layer on a circular cylinder separates at about $\theta = 80^\circ$. Compare this observation with your predicted result and explain why there is a difference.

Hint

$$\int \sin^5 x \, dx = -\frac{1}{5} \sin^4 x \cos x + \frac{4}{15} \cos^3 x - \frac{4}{5} \cos x$$



$$\therefore \theta^+(x_0) = 0.45 U_e^{-6}(x_0) \nu \int_0^{x_0} U_e^5(x) \, dx$$

$$\therefore \theta^+(\theta) = 0.45 U_e^{-6}(\theta) \nu \int_0^\theta U_e^5(\theta) r \, d\theta \quad \text{in this case}$$

$$\text{where } \int_0^\theta U_e^5 r \, d\theta = (2U_\infty)^5 r \int_0^\theta \sin^5 \theta \, d\theta$$

$$= (2U_\infty)^5 r \left[-\frac{1}{5} \sin^4 \theta \cos \theta + \frac{4}{15} \cos^3 \theta - \frac{4}{5} \cos \theta \right]_0^\theta$$

$$= (2U_\infty)^5 r \left(-\frac{1}{5} \sin^4 \theta \cos \theta + \frac{4}{15} \cos^3 \theta - \frac{4}{5} \cos \theta + \frac{8}{15} \right)$$

$$\therefore \theta^+(\theta) = \frac{9}{40} \frac{\nu r}{U_\infty} \left(-\frac{1}{5} \sin^4 \theta \cos \theta + \frac{4}{15} \sin^6 \theta \cos^3 \theta - \frac{4}{5} \sin^6 \theta \cos \theta + \frac{8}{15} \sin^6 \theta \right)$$

$$\frac{dU_e}{r d\theta} = \frac{d}{d\theta} \frac{1}{r} (2U_\infty \sin \theta) = \frac{1}{r} 2U_\infty \cos \theta$$

$$\therefore m = -\frac{\theta^+}{\nu} \frac{dU_e}{dx}$$

$$= -\frac{9}{20} \left(-\frac{1}{5} \sin^4 \theta \cos^2 \theta + \frac{4}{15} \sin^6 \theta \cos^4 \theta - \frac{4}{5} \sin^6 \theta \cos^2 \theta + \frac{8}{15} \sin^6 \theta \cos \theta \right)$$

$$= -\frac{9}{20} \left(-\frac{1}{5} \frac{\cos^2 \theta}{\sin^4 \theta} + \frac{4}{15} \frac{\cos^4 \theta}{\sin^6 \theta} - \frac{4}{5} \frac{\cos^2 \theta}{\sin^6 \theta} + \frac{8}{15} \frac{\cos \theta}{\sin^6 \theta} \right)$$

At separation point, $m = 0.082$

Use iterative method by MATLAB, as shown

```
close all
clear
clc

theta = 0; % Initial angle
result = 1; % Pre-defined error

while abs(result - 0.082) > 0.01

    theta = theta + 0.001; % Increase angle by 0.1 rad

    result = - (9 / 20) * (-(1 / 5) * (cos(theta)) ^ 2 / (sin(theta)) ^ 2) + ((4 / 15) * (cos(theta)) ^ 4 / (sin(theta)) ^ 6) ...
        - ((4 / 5) * (cos(theta)) ^ 2 / (sin(theta)) ^ 6) + ((8 / 15) * cos(theta) / (sin(theta)) ^ 6));

end

theta = theta * 180 / pi;
```

$$\therefore \theta_{sep} = 101.4^\circ$$

- In reality, at low Re , $\theta_{sep} = 80^\circ < 101.4^\circ$

This is because Thwaite's method doesn't account von Kármán vortex street

4. Calculate the skin friction coefficient and the shape factor at $\theta = 90^\circ$ on the circular cylinder given in the previous question for a Reynolds number, based on cylinder diameter of 8×10^4 .

ANS 0.0032, 2.61

$$\text{At } \theta = 90^\circ, \quad m = 0, \quad \theta'(\theta) = \frac{3}{25} \frac{vr}{U_\infty}$$

$$\therefore U_e = 2U_\infty$$

$$\theta = \frac{\sqrt{3}}{5} \sqrt{vr}$$

$$\therefore l = 0.22 = \frac{\theta}{U_e} \left(\frac{\partial U}{\partial y} \right)_w$$

$$H = 2.61$$

$$\therefore \left(\frac{\partial U}{\partial y} \right)_w = 0.22 \frac{U_e}{\theta}$$

$$\therefore C_f = 2 \frac{\tau_w}{\rho U_e^2} = 2 \frac{\mu \left(\frac{\partial U}{\partial y} \right)_w}{\rho U_e^2}$$

$$= 0.44 \frac{\mu}{\rho U_e \theta} = 0.44 \frac{\mu}{\rho U_\infty \theta}$$

$$= 0.44 \frac{\nu}{U_\infty} \frac{5}{\sqrt{3}} \frac{1}{\sqrt{vr}} = \frac{11}{10\sqrt{3}} \sqrt{\frac{\nu}{U_\infty r}}$$

$$\therefore Re = \frac{\rho U_\infty c}{\mu} = \frac{U_\infty c}{\nu} = \frac{2U_\infty r}{\nu} = 8 \times 10^4$$

$$\therefore C_f = \frac{11}{10\sqrt{3}} \sqrt{\frac{2}{Re}} = 3.175 \times 10^{-3}$$

5. A turbulent boundary layer, measured in standard atmospheric conditions, is found to be 6 cms thick. The velocity is 30 m/s at the edge of the layer, and 20m/s at 0.6 cms from the surface. From these measurements determine the local skin friction coefficient, assuming that, in the appropriate region the distribution of velocity in the boundary layer is given by

$$U / U_{\tau} = 5.62 \log_{10} (U_{\tau} y / \nu) + 5.2.$$

It is found that, over most of the boundary layer thickness, the measured velocity profile is satisfactorily fitted by the relation $U/U_e = (y/\delta)^{1/6}$. Use this to find the value of the local skin friction coefficient given by the Ludwig-Tillmann semi-empirical relation:

$$C_f = 0.246 \times 10^{-0.678H} Re_0^{-0.268}.$$

All symbols have their usual meaning.

• $\delta = 6 \text{ cm}, U_e = 30 \text{ m/s}, U|_{y=0.6 \text{ cm}} = 20 \text{ m/s}$

$$\frac{U_e}{U_{\tau}} = \frac{1}{K} \ln \left(\frac{\delta U_{\tau}}{y} \right) + C + \frac{2\pi(x)}{K}$$

$$\frac{U}{U_{\tau}} = \frac{1}{K} \ln \left(\frac{y U_{\tau}}{y} \right) + C$$

$$\therefore \frac{U_e - U}{U_{\tau}} = \frac{1}{K} [\ln \left(\frac{\delta}{y} \right) + 2\pi(x)] = \frac{1}{K} [-\ln \left(\frac{y}{\delta} \right) + 2\pi(x)]$$

$$\approx \frac{2\pi(x)}{K} \text{ for } \frac{y}{\delta} \rightarrow 1$$

just revising
(not relate to
this question)

when $y = 0.6 \text{ cm} = 0.006 \text{ m},$

$$\frac{20}{U_{\tau}} = 5.62 \log_{10} \left(\frac{0.006 U_{\tau}}{y} \right) + 5.2$$

At standard atmospheric conditions, $\mu = 1.82 \times 10^{-5} \text{ kg/ms}, \rho = 1.19 \text{ kg/m}^3$

$$\therefore \nu = \frac{\mu}{\rho} = 1.529 \times 10^{-5}$$

Use iterative method by MATLAB, as shown

```
close all
clear
clc

u = 0;

v = 1.82e-5 / 1.19; % dynamic viscosity

result = 1; % Pre-defined error

while abs(result) > 0.01

    u = u + 0.0000001; % Increase angle by 0.1 rad

    result = 5.62 * log10(0.006 * u / v) + 5.2 - 20 / u;

end
```

$$\therefore u_{\tau} = 1.0096 \text{ (m/s)} = \sqrt{\frac{\tau_w}{\rho}}$$

$$\therefore \tau_w = u_{\tau}^2 \rho = 1.2130 \text{ (N)}$$

$$\therefore C_f = 2 \frac{\tau_w}{\rho U_e^2} = 2.265 \times 10^{-3}$$

$$\cdot \frac{U}{U_e} = \left(\frac{y}{\delta}\right)^{\frac{1}{2}}$$

$$\therefore \frac{\theta}{\delta} = \int_0^1 \left(\frac{y}{\delta}\right)^{\frac{1}{2}} \left[1 - \left(\frac{y}{\delta}\right)^{\frac{1}{2}}\right] d\left(\frac{y}{\delta}\right)$$

$$= \int_0^1 \left(\frac{y}{\delta}\right)^{\frac{1}{2}} - \left(\frac{y}{\delta}\right)^{\frac{3}{2}} d\left(\frac{y}{\delta}\right)$$

$$= \left[\frac{6}{7} \left(\frac{y}{\delta}\right)^{\frac{7}{2}} - \frac{3}{4} \left(\frac{y}{\delta}\right)^{\frac{5}{2}}\right]_0^1$$

$$= \frac{3}{28}$$

$$\text{and } \frac{\delta^*}{\delta} = \int_0^1 1 - \left(\frac{y}{\delta}\right)^{\frac{1}{2}} d\left(\frac{y}{\delta}\right)$$

$$= \left[\frac{y}{\delta} - \frac{6}{7} \left(\frac{y}{\delta}\right)^{\frac{7}{2}}\right]_0^1$$

$$= \frac{1}{7}$$

$$\therefore H = \frac{\delta^*}{\theta} = \frac{4}{3}$$

$$Re_{\theta} = \frac{\rho U_e \theta}{\mu} = \boxed{117692.3077} \quad 12609.89$$

$$\therefore C_f = \boxed{1.3427 \times 10^{-3}} \quad 2.443 \times 10^{-3}$$